INTE 22293 - Software Architecture & Process Models

In-class Activity 01
Reverse Engineering a Real-World Software System

Duration: 3 Hours

- **Preparation:** 2 Hours
- **Presentation:** 5 Minutes per Group

Learning Outcomes

By completing this activity, students will be able to:

- Analyze an existing software system from an architectural perspective.
- Identify stakeholders and major functional requirements.
- Think critically about how large-scale software systems are designed.
- Explain architectural decisions using logical reasoning.

Background

Every successful software product is built on a well-designed software architecture. Although companies rarely publish their complete architecture, software engineers and architects often analyze existing products to understand how they might have been designed.

In this activity, your team will **reverse engineer** a popular software system. This does **not** mean finding its actual source code or exact architecture. Instead, you will use observation, research, logical reasoning, and your software engineering knowledge to infer how the system may be structured.

Activity Instructions

Work in groups of **4–6 students**.

Each group should select **one software system** from the suggested list below or choose another system you preferred.

During the next **2 hours**, your team will analyze the selected system and prepare a **5-minute presentation**.

You may use the Internet or AI tools to gather publicly available information, but the presentation and explanations should be your own work.

Suggested Software Systems

Transportation & Mobility

- Uber
- PickMe
- Uber Eats

Social Media

- Instagram
- Facebook
- TikTok
- LinkedIn

Messaging & Communication

- WhatsApp
- Discord
- Microsoft Teams
- Zoom

Video & Entertainment

- YouTube
- Netflix
- Spotify

E-Commerce

- Amazon
- Daraz
- eBay
- Alibaba

Finance

- PayPal
- Wise
- Revolut

Travel

- Airbnb
- Booking.com

Artificial Intelligence

- ChatGPT
- Google Gemini
- Microsoft Copilot

Education

- Moodle
- Coursera
- Google Classroom

Productivity

- Google Drive
- Google Docs
- Notion
- Trello

Your Task

Your presentation should cover the following sections.

1. System Overview

Briefly introduce the system.

Include:

- Purpose of the system
- Target users
- Core services provided
- Why you selected this system

2. Stakeholders

Identify the main stakeholders.

Example:

- End users
- Administrators
- Business owners
- Delivery partners
- Merchants
- Drivers
- Content creators
- Advertisers

3. Functional Requirements

List the major functions provided by the system.

Example:

For WhatsApp

- User Registration
- Login

- Send Messages
- Voice Calls
- Video Calls
- File Sharing
- Group Chats

4. Non-Functional Requirements

Discuss important quality attributes.

Consider:

- Performance
- Security
- Scalability
- Availability
- Reliability
- Maintainability
- Usability
- Privacy

Explain **why** each is important for your selected system.

5. High-Level Architecture

Draw a high-level architecture diagram showing the major components.

Include appropriate components such as:

- Mobile App / Web Application
- API Gateway
- Backend Services
- Authentication Service
- Database
- Cache

- File Storage
- Notification Service
- Payment Gateway (if applicable)
- Third-party APIs
- Cloud Infrastructure

Your diagram does **not** need to represent actual architecture. It should represent a logical design based on your understanding.

6. Data Flow

Choose one common user scenario and explain how data flows through the system.

Examples:

- Sending a WhatsApp message
- Booking an Uber ride
- Watching a Netflix movie
- Ordering food
- Uploading a photo to Instagram

Describe the sequence of interactions between the user and system components.

7. Scalability Considerations

Assume the system suddenly grows from **100,000 users to 100 million users**.

Discuss how architecture could support this growth.

Possible ideas:

- Load balancing
- Multiple application servers
- Database replication
- Caching
- Content Delivery Network (CDN)
- Microservices

- Cloud auto-scaling
- Queue-based processing

8. Security Considerations

Identify possible security requirements.

Examples:

- Authentication
- Authorization
- Encryption
- Secure payment processing
- Multi-factor authentication
- Data privacy
- Protection against cyber attacks
- Backup and disaster recovery

9. Technology Stack

Based on your understanding, suggest technologies that could be used.

Examples:

Frontend

- React
- Angular
- Flutter

Backend

- Spring Boot
- Node.js
- .NET
- Django

Database

- PostgreSQL
- MySQL
- MongoDB

Infrastructure

- AWS
- Azure
- Google Cloud

These do not have to match the real technologies used.

Deliverables

Each group should prepare:

- Presentation slides
- One architecture diagram
- One data flow diagram

Presentation Guidelines

Each group will have:

- **5 Minutes** for presentation

Group number, member names and student numbers should be included in the presentation.

Assessment Rubric

Criteria	Marks
Understanding of the Selected System	10
Functional & Non-functional Requirements	15
Architecture Diagram	20
Data Flow Explanation	10
Scalability & Security Analysis	15
Technology Stack Justification	10
Creativity and Critical Thinking	10
Presentation Quality & Teamwork	10
Total	**100**

Important Notes

- Creativity and critical thinking are encouraged.
- Focus on explaining **why** architectural decisions make sense.
- Keep your presentation concise, visual, and easy to understand.